

Digital twins and artificial intelligence in transportation infrastructure: Classification, application, and future research directions

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ABSTRACT

Artificial Intelligence (AI) technology is extensively applied in all walks of life with continuous acceleration of the construction of smart cities. The current research status of intelligent development of transportation infrastructure is classified and predicted oriented to smart cities, which is mainly based on space type, function type, and facility use. The classification is mainly between residents and vehicles, between and vehicles, and between vehicles and transportation systems. The related scholars analyzed that Digital Twins (DTs) technology and AI technology show significant advantages in the classification of transportation infrastructure and the management of transportation spatial information network. The intelligent development of transportation infrastructure is analyzed, and it is predicted in functional design, intelligent development, and effective integration with new media, aiming to provide a reference for the subsequent intelligent development and construction of transportation infrastructure in smart cities.

1. Introduction

With the development of Smart City, technologies such as the Internet of Things, Cloud Computing, Artificial Intelligence (AI), and Digital Twins (DTs) have been widely applied in all walks of life. DTs technology is the core to realize Cyber-Physical System (CPS). It makes full use of data (such as physical model, sensor update, and operation history), integrates multi-disciplinary, multi-physical, multi-scale, multi-probability simulation process, completes the mapping in virtual space, and reflects the full life cycle process of the corresponding physical equipment. DTs technology provides new ideas for vehicle development, intelligent operation and maintenance of infrastructure, virtual test of unmanned driving, and live traffic analysis. It effectively solves the complex urban traffic congestion and slow operation in the transportation field.

Transportation is one of the important components of smart city construction, and construction of transportation infrastructure is its basic link. In China, the construction of transportation infrastructure is moving towards the new situation of digitalization, new media, and networking. The construction of good transportation infrastructure not only improves the people's daily life, but also further enhances their esthetic perception. Therefore, the design of transportation infrastructure needs to integrate multiple data sources, pay attention to the macro composition and data analysis of the platform, and analyze the functional requirements in detail. The use of DTs technology can effectively solve the existing problems in the transportation field. The intelligent transportation systems

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can effectively manage road network traffic and realize intelligent changes in traffic monitoring in smart city. Realizing the digitization of transportation infrastructure requires the integration of multiple data sources, combining building information modeling (BIM), geographic information system (GIS) data, and aerial photography data of traffic conditions to build a digital transportation infrastructure platform with large storage capacity, high storage efficiency, simple structure, and stable operating environment [11]. Therefore, the establishment of a real-time, accurate, efficient, and comprehensive transportation management system can effectively promote the active two-way information interaction between vehicles and roads and improve smart traffic efficiency.

Nowadays, intelligent technologies are widely used in all walks of life, the intelligent construction of transportation infrastructure is essential, and intelligent transportation systems are emerged. The predecessor of intelligent traffic system (ITS) is an intelligent vehicle highway system (IVHS) [2]. The ITS effectively and comprehensively applies advanced information technology, data communication technology, sensor technology, electronic control technology, and computer technology, so it establishes a large-scale, all-round function, real-time, accurate and efficient integrated transportation management system. Intelligent transportation systems can improve transportation efficiency, alleviate traffic congestion, increase traffic capacity of the road network, reduce traffic accidents, lower energy consumption, and reduce environmental pollution through the harmonious and close cooperation of people, vehicles, and roads.

As the basic guarantee for the real-time dynamic information interaction of people, vehicles, roads, and the environment in the intelligent transportation system, information and communication technology is an important support for the development of an efficient, safe, and intelligent transportation system. Lv et al. applied wireless communication technology to intelligent transportation systems, which could not only realize the collection of the vehicle's own state information, the perception of driving environment information, and the planning and tracking of vehicle trajectories, but also effectively change the path during congestion [3]. Ambra and Macharis investigated the synchro modal road transport mechanism to support the real-time optimization and integration of different transport modes and infrastructures to cause the transformation of transport modes from roads to inland waterways and railways. The DTs technology was applied to establish a symbiotic relationship between the virtual environment and the physical environment, helping to make mode selection, order synchronization, and enable the cost more acceptable and the lead time more dynamic and flexible [4]. Therefore, ITS is becoming the main development direction of future transportation. The ubiquitous communication system can be fully covered in the transportation infrastructure, realizing the real-time information interaction of vehicle-road/vehicle-vehicle, and establishing the intelligent integration of people, vehicles, roads, and the environment. In addition, using the DTs technology in the ITS can map the road network traffic infrastructure in the real physical space to the virtual space, and comprehensively reflect the road network traffic infrastructure. The vehicular ad-hoc network (VANET) is incorporated into the modern transportation infrastructure and autonomous vehicles are integrated into the ITS, to achieve efficient communication between vehicles and vehicles (V2V communication) or between vehicles and infrastructure (V2I communication), improve driving safety, and avoid traffic congestion and better coordinate the traffic network [5]. In addition, application of DTs technology in transportation infrastructure can make the system realize the overall infrastructure construction instead of a single infrastructure component construction, and finally realize the transparent interconnection of the road network transportation infrastructure after integration. With the development of 4G, 5G, and even more advanced communication technologies and coupled with end-side-cloud collaborative computing, the real-time performance of DTs can be improved, and it is even possible to model unknown areas in real time without HD-Map (high-definition map). Secondly, by improving behavior simulation and prediction algorithms, the behavior prediction can be more accurate.

In this work, the current research status of transportation infrastructure in the process of smart city construction is classified, and the development trend of the smart city transportation is analyzed and predicted, to provide a reference for the intelligent development direction of the transportation under urbanization. In the chapter 2, the application classification of current transportation infrastructure is analyzed in the context of smart cities, and the intelligent traffic accident detection is discussed based on AI and DTs. In the chapter 3, the future development direction of intelligent transportation is predicted relying on modern technologies such as the Internet of Things and AI, to promote the efficient operation of transportation. Compared with the existing research, this work more systematically and comprehensively integrates the application of AI and DTs technology in smart transportation, and explains the specific application examples, which is of great significance for further clarification of the future development direction of smart transportation.

2. Transportation infrastructure classification and traffic accident detection

2.1. Intelligent classification method for transportation infrastructure

In the construction of a smart city, the construction of public infrastructure is its basic link, which is classified into three perspectives. Based on the space type, it can be classified into transportation space facilities, urban park facilities, commercial space facilities, educational space facilities, and residential community space facilities. Based on the function type, it can be classified into educational facilities, medical and health facilities, cultural and recreational facilities, sports facilities, social welfare and security facilities, administrative management and community service facilities, commercial facilities, and municipal public facilities. Based on the facility use, it can be classified into urban public facilities hardware systems and urban public facilities software systems. Transportation infrastructures are public infrastructures of a city, which are mainly classified by spatial type to ensure the convenience of public travel in the transportation network.

According to the composition of national transportation system in China, the transportation infrastructure is classified into five categories: railway transportation infrastructure, road transportation infrastructure, waterway transportation infrastructure, air

transportation infrastructure, and pipeline transportation infrastructure. According to the characteristics of transportation routes, it covers three categories: land transportation infrastructure, air transportation infrastructure, and waterway transportation infrastructure. According to the mode of transportation, four categories can be defined: rail transportation, pipeline transportation, waterway transportation, and road transportation. According to the function of basic transportation facilities there are three types: basic transportation facilities between urban residents and vehicles, basic transportation facilities between vehicles, and basic transportation facilities between vehicles and transportation systems.

With the advancement of modern AI technology, the construction of traditional transportation infrastructure is developing toward intelligence. In addition, advanced fields such as mobile Internet, Machine Learning, Big Data, and DTs are gradually integrated with transportation infrastructure construction. In this work, the application classification of transportation infrastructure is mainly analyzed from the perspective of function.

Transportation infrastructure occupies a large proportion in cities, and has various functions and types, whose final goal is to provide convenience for urban residents to travel. It can be categorized into three aspects according to the function. The first refers to the related public facilities between urban residents and transportation, such as zebra crossings, bus stations, and bicycle rental stations; the second refers to the related public facilities between transportation, such as traffic lights, green belts, traffic signs and streetlights; and the third refers to public facilities associated with vehicles and the transportation system, such as roadside units, edge nodes, and sensor terminals. The classification of transportation infrastructure is shown in Fig. 1.

2.2. Intelligent traffic accident detection based on artificial intelligence and digital twins

DTs technology is a virtual mapping of a physical entity. With the help of historical data, real-time data, and algorithm models, it is a technical means to simulate, verify, predict, and control the entire life cycle process of a physical entity. Applied in road traffic, DTs technology can not only realize the virtualized mapping of physical entities, but also realize dynamic monitoring of the life cycle of road infrastructure and accurate restoration of traffic participants on the road using a variety of sensors and network communication

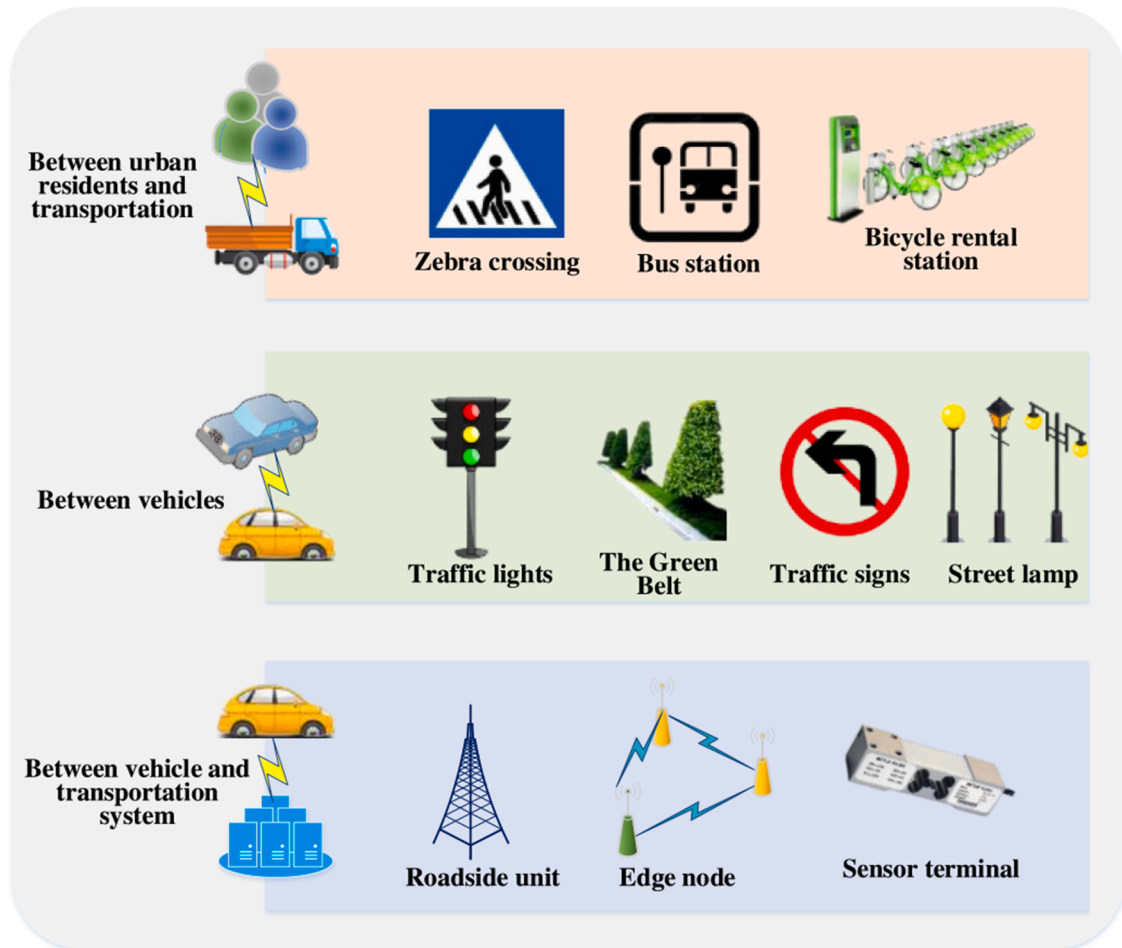


Fig. 1. Schematic diagram of the classification of transportation infrastructure.

technologies. In addition, it can judge and predict possible risks based on traffic behavior, analyze road traffic conditions based on traffic conditions, and provide accurate basis for road traffic diagnosis and decisions. In the actual construction of smart transportation, based on the digitization of infrastructure and multi-source heterogeneous physical perception, the data is uploaded to the computing platform with twin function by using a variety of low-latency network combinations as channels for modeling, reconstruction, and accurate perception, to apply digital and structured data to various transportation services.

The core of the DTs is to convert physical roads, infrastructures, and traffic targets into numbers with characteristic information, and thus into language for machines to automatically read and recognize. Under this basic premise, it can obtain the state process of the whole life cycle of the roads and vehicles, directly provide the traffic participation target including the position, speed, angle, contour, and type to the computing unit for reading, and automatically determine the target behavior. Different from traditional video surveillance, the multi-dimensional presentation of DTs is not affected by light conditions, providing the most intuitive and comprehensive understanding of real-time traffic status. In addition, it can flexibly switch any angle of view and quickly check the occurrence of traffic incidents from the traffic situation of the road network to the behavior of micro-vehicles.

Traffic congestion caused by traffic accidents seriously affects the normal travel of the public, which may be alleviated by studying accurate and effective methods. The traffic DTs is an evolving traffic system digital archive that should help optimize the processes in traffic. Therefore, the traffic DTs should rely on high-precision traffic detectors' data. After a traffic accident, the DTs system and traffic data can be combined to restore the entire process of the traffic accident, explore the accident situation at that time and whether each step of the response is right and whether there is room for improvement, etc. The combination of big data, machine learning, and advanced sensing technology in the transportation infrastructure can detect vehicle damage. Wang et al. found that smart internet of vehicles (SIoV) can realize real-time content dissemination in urban traffic management systems. The research outlined several promising research areas of SIoV traffic management, and designed a framework based on crowd perception to provide timely responses to traffic management in heterogeneous SIoV. In this research, they analyzed the vehicles based on device to device (D2D) communication to dynamically adjust their behaviors according to network conditions by integrating trajectory and topology

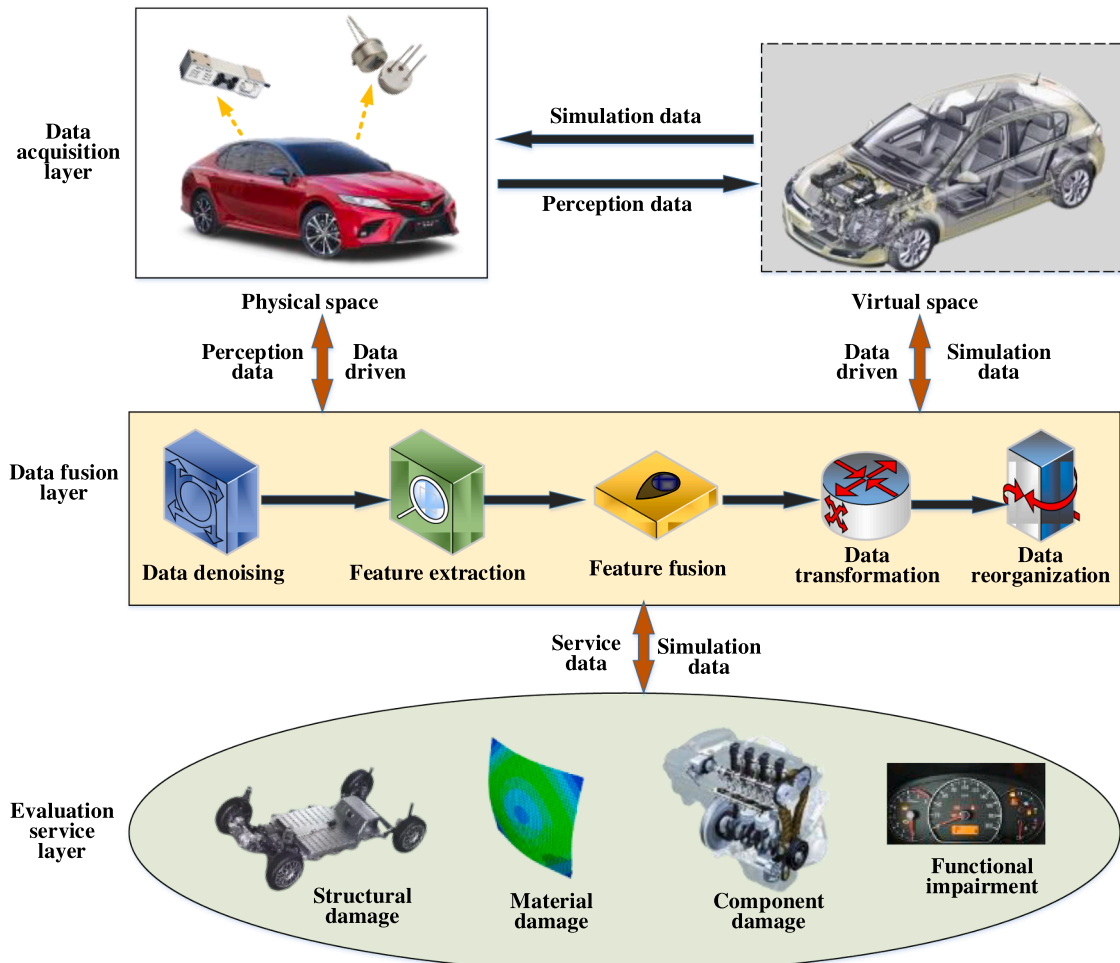


Fig. 2. Schematic diagram of traffic vehicle damage assessment based on DTs technology and AI.

information. Finally, the performance evaluation was carried out based on the actual trajectory analysis, which verifies the effectiveness of the framework based on crowd perception [6]. Ke et al. proposed a road congestion detection method based on multi-dimensional visual features and convolutional neural network (CNN), aiming at the problems that traditional traffic congestion detection methods are difficult to use, costly, and may cause damage to road. The gray level co-occurrence matrix was adopted to detect the density of foreground objects, the Lucas-Kanade optical flow pyramid was applied to detect the speed of moving objects, the Gaussian mixture model was combined to model the background, and finally the CNN was introduced to accurately predict the state of traffic congestion [7].

The studies of the above-mentioned scholars have proved that the transportation infrastructure has played a particularly important role in the detection of traffic accident damage. The DTs technology in traffic is composed of three parts: the physical traffic entity in the real world, the virtual traffic facility model, and the associated data and views that link the entity-virtual model together. The DTs technology in transportation is divided into a physical system and a virtual system. The physical system provides the foundation for development of the virtual system, while the virtual system is responsible for the simulation, control, and optimization strategies of the service system. The comprehensive service platform can response to the needs of physical and virtual systems. As one of the important components in the transportation system, vehicle is a complex system covering materials science, mechanical design, and control science. Under diversified working conditions, abnormal conditions may occur in the shell material, internal structure, and functions of the vehicle during the working process. Different damage sources such as collisions, dust, and external attacks, may have varying degrees of impact on vehicles. At this stage, in the traffic system the vehicle damage assessment generally adopts physical simulation, but this method has high cost, low precision and low reliability. After the introduction of DTs technology, the materials, structures, and components of vehicles in real space can be mapped in the virtual system for testing and evaluation, to evaluate the performance of the vehicle further comprehensively. The damage evaluation of traffic vehicles based on DTs technology and AI is shown in Fig. 2.

The DTs for vehicle damage assessment is to digitally create virtual entities of the damaged physical entities (including the damaged function and parts of the vehicle) and use real-time data and traffic algorithm models to simulate, verify, and predict all types of traffic accidents. Vehicle damage assessment based on DTs and AI technology is the real-time information interaction and two-way real mapping between physical and virtual vehicles, to realize the full services in physical space and virtual space. The integration and fusion of data will ultimately provide a reliable assessment service for the damage of the vehicle. The DTs vehicle is composed of five parts: physical vehicle, virtual vehicle, twins' data, dynamic real-time connection, and service. The physical vehicle is composed of the vehicle itself and its sensing system. The sensing system collects damage-related data from the vehicle entity and transmits it to the virtual space to support high-precision simulation of the virtual vehicle. The physical data is merged with virtual data, to extract the characteristics of the anti-damage performance of the virtual vehicle and assist the construction of the model. The virtual model is a high-fidelity model that contains a multi-dimensional fusion of geometric model, physical model, behavioral model, and regular model, which can truly depict and map the state of physical vehicles. Dynamic real-time connection is driven by modern information transmission technology, through efficient, fast and accurate detection technology, to achieve real-time information interaction among physical vehicles, virtual vehicles, and services. Vehicle anti-damage performance evaluation requires integration of historical data and real-time data, analyzing comprehensively the multi-dimensional characteristics of the materials, structure, components, and functions. As the core technology for the creation and operation of the traffic DTs, the traffic vehicle simulation technology is the basis for the data interaction and integration. On this basis, the traffic DTs integrates other new technologies, to ensure its fidelity, real-time, and closed loop. Although DTs technology is the advanced trend of smart transportation, there is still a certain gap from the real DTs transportation system with global management, simultaneous visualization, and virtual-real interaction. However, driven by 5G technological changes and demand upgrades, DTs technology has provided new ideas, new approaches, and new concepts for smart transportation, which will continue to develop in the future and eventually form a complete technical operation system.

3. Prospects for the intelligent development of transportation infrastructure

3.1. The development of intelligent transportation based on the internet of things

The intelligent transportation industry involves many contents, and needs the cooperation of many fields, such as telecom operators, satellite navigation, big data, and vehicle terminals. The car covered by the smart transportation construction is no longer a simple means of transportation, nor is it a source of increasingly severe urban traffic congestion and serious air pollution, but a comprehensive transportation link based on the Internet of Things, Big Data, new energy, and autonomous driving. As the main carrying medium of transportation, modern roads need to be combined with new technologies to form a new type of traffic carrying system with information interaction in addition to scientific construction and comprehensive application of new materials. In the construction of smart transportation, the environment is no longer a simple natural concept, but a comprehensive medium including technology application, multi-dimensional perception and business carrying. For the construction of smart city, the intelligence of transportation is one core, and the difficulties and pain point it faces also affect the construction process of the entire smart city. The main bottlenecks can be roughly classified into "three difficulties": it is difficult to exchange information; it is difficult to analyze traffic big data; and it is difficult to manage account.

As an integral part of urban space, urban transportation facilities correspond to different groups of users of urban buildings and streets. Therefore, in the subsequent design and improvement of urban transportation facilities, the entire space environment should be considered comprehensively and systematically, starting from its functions, shapes, and materials, and merging with the surrounding environment to achieve a rich space environment.

A transportation infrastructure needs to satisfy its own basic functionality. Based on the existing transportation infrastructure and

transportation needs, a facility that meets the requirements of users to the greatest extent is designed. Nowadays, the transportation infrastructure is developing toward the direction of intelligence to ensure the safety and smoothness of traffic. In intelligent traffic monitoring, various sensor technologies are mainly used to monitor vehicle driving conditions and road congestion, prevent problems that may occur, and strengthen the monitoring of accident-prone areas. The combination of transportation infrastructure and Internet of Things technology collects and processes on-site real-time data, effectively alleviating traffic jams.

With the rapid development of Internet technology, the dissemination of information is not only limited to newspapers and books but is spread through New Media such as the Internet. New Media uses its interactivity to disseminate and respond to traffic infrastructure information. Smart cities collect real-time data from Internet of Things devices in the urban environment, and use these data and 3D models for analysis, testing, optimization, and predictive verification. Based on the characteristics of transportation infrastructure servers, the combination of New Media and transportation infrastructure is bound to become the future development trend.

3.2. Prospects and applications of the development of intelligent transportation infrastructure

When it comes to the prospects for the intelligent development of transportation infrastructure, the analysis is carried out from the perspective of functional design, intelligent development design, and effective integration with New Media. The specific prospective application of transportation infrastructure is shown in Fig. 3.

3.2.1. Prospective application of functional design of transportation infrastructure

The functional design of road traffic infrastructure can greatly enhance its functionality, and trash cans is one of the indispensable infrastructures in any area. To actively respond to the domestic garbage classification policy, the types of garbage have increased from the original two types of recyclable garbage and non-recyclable garbage to four types: recyclable garbage, kitchen waste, hazardous garbage, and other garbage. Smart cities integrate various solid waste monitoring systems, video monitoring equipment data, etc., based on GIS, to monitor the quantity and distribution of solid and semi-solid waste in real time, and to adjust the number of trash cans placed on different roads according to the amount of garbage. Because there are still some problems in the actual garbage classification process, many scholars have conducted research on the intelligent improvement of garbage classification in transportation infrastructure. Boualouache et al. proposed a complete and efficient pseudonym change and management framework to classify the private location of vehicles and self-organizing networks, which improved the functionality of roadside gas stations and trash cans [8]. Wang and Wang developed a new DTs-based system for waste electrical and electronic equipment (WEEE) recycling to support manufacturing/remanufacturing operations throughout the product life cycle (from design to recycling) and established a data model that complied with international standards to support WEEE recovery service with high data interoperability. The feasibility of the system proposed has been verified and evaluated during the implementation process in cloud and CPS [9].

After the functional improvements in the transportation infrastructure are analyzed, it is found that the establishment of a long-term waste disposal system of classified release, classified collection, classified transportation, and classified treatment can achieve

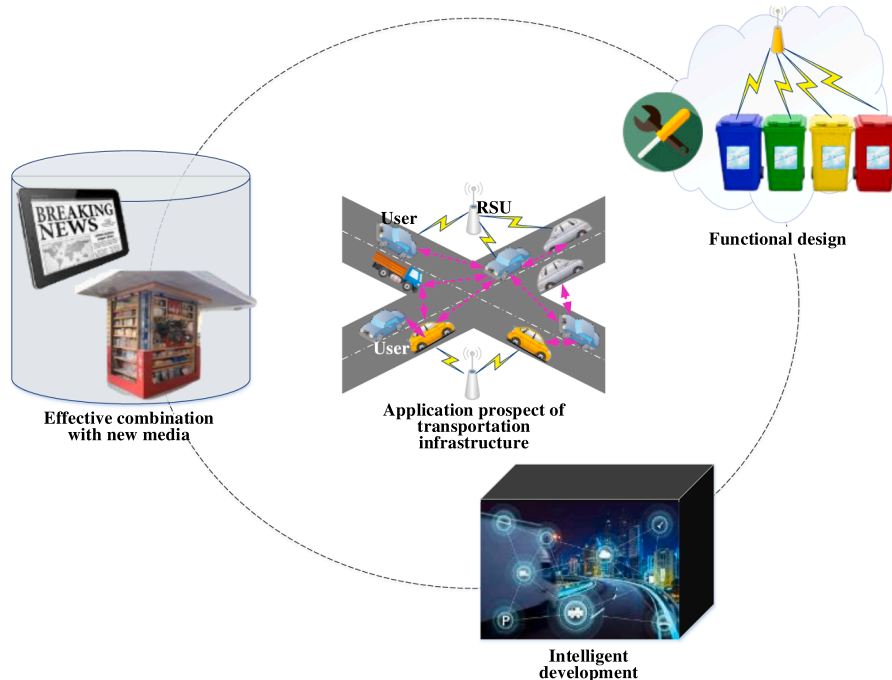


Fig. 3. Schematic diagram of the application of intelligent transportation infrastructure prospects.

waste reduction, resource utilization, and harmlessness. The establishment of DTs-based intelligent environmental traffic visualization system can reasonably allocate the resources of transportation environmental facilities, and then recycle part of the garbage through the reasonable classification of the garbage in the transportation infrastructure, fully excavating its potential value. After removal of plastic waste, it can reduce the carbon dioxide emissions in the waste incineration process, which has important environmental significance. The functional improvement of trash cans in transportation infrastructure is shown in Fig. 4.

When the functionality of trash cans in transportation infrastructure is enhanced, intelligent recognition of images combined with neural network technology can achieve correct allocation and arrange of trash to be able to effectively classify various types of trash. Functional improvements mainly include image acquisition, image processing, feature extraction, and image recognition and classification. In the image acquisition, some typical recyclable domestic garbage in the major garbage recycling transfer stations is photographed, based on which the garbage image feature database is constructed. Combined with ImageNet data set, the garbage image feature database is classified, which can be divided into waste glass, waste plastic, waste metal, etc. All kinds of commodity packaging materials have bar codes, so it is necessary to build an identification library to distinguish which category of garbage each bar code belongs to. The identification library classifies the garbage into paper, glass, metal, plastics, and cloth. Then, the garbage classification identification library, global positioning system (GPS), upper computer, lower computer, and service network platform realize the intelligent classification of trash cans based on bar code recognition through programming, mechanical assembly, and network series. The intelligent classification trash cans provide users with accurate information and guidance for garbage classification. The smart city uses professional industry model analysis algorithms to visually analyze and judge factors such as the comprehensive utilization rate of industrial solid waste, the centralized disposal rate of medical waste, the harmless treatment rate of domestic waste, and the coverage rate of domestic waste classification, to assist managers in a comprehensive grasp. Therefore, the intensity of solid waste disposal will be increased, and the quality of the urban traffic environment will be greatly improved.

3.2.2. Prospect and application of intelligent design of transportation infrastructure

The intelligent design of transportation infrastructure is mainly reflected in the use of sensor technology to strengthen monitoring, the use of embedded technology and Internet of Things technology to collect and process on-site real-time data, and the use of DTs and AI technology to create digital cities, to provide users with intelligent services such as urban vehicle location and road safety prediction. Based on the global perception technology of DTs, traffic facilities are endowed with intelligent perception functions to remotely monitor urban traffic flow and road congestion.

When the embedded technology and Internet of Things technology are adopted to collect and process on-site real-time data, many scholars have conducted research on related intelligent transportation infrastructure. As the foundation of transportation vehicle infrastructure, the development of the intelligent cockpit will experience the transformation from physical buttons to digital touch screens to voice control to multi-modal interaction. Different from the previous interaction modes, multi-mode interaction has undergone qualitative changes in the input and output modes. Voice can liberate hands and eyes of people, reduce the operation risk of driving scenes, and improve safety. The development trend of intelligent driving cockpit in the future is active, multi-mode, and multi-scene interaction design [10].

The analysis of the above-mentioned scholars' research suggests that the real-time data processing of traffic by the Internet of Things technology has become a new trend in the current intelligent development. With the continuous increase of urban rail transit lines, the demand for new services continues to increase, and the operational production data has exploded. Based on technologies such as DTs, GPS satellite positioning, and smart sensing, smart cities can break through traditional models, realize intelligent monitoring of abnormal conditions of the city's ground and underground space, and guide the construction and operation management of DTs-based cities. For example, a 3D urban spatial model is established through the urban information model to realize comprehensive 3D visual management of urban traffic planning, construction, and operation, support urban traffic simulation, analysis, and verification, and realize intelligent supervision of urban transportation. This work is based on the current research, introduces big data analysis technology, and builds a Hadoop big data platform, so that the application of real-time data processing and analysis technology in rail transit is continuously improved. The real-time data processing framework of rail transit based on the Internet of Things and big data analysis is shown in Fig. 5.

Fig. 5 shows the flow chart of real-time data processing of rail transit using the Internet of Things and Big Data analysis technology.

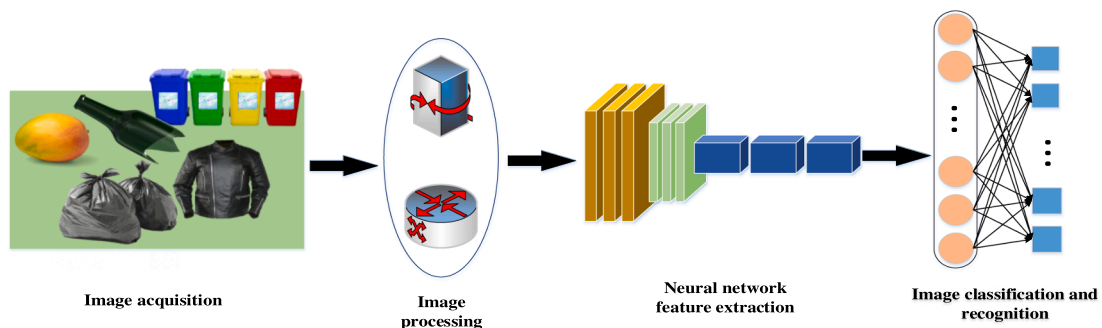


Fig. 4. Schematic diagram of functional improvement of trash cans in transportation infrastructure.

Real-time data of urban rail transit mainly includes on-site information such as passenger flow of lines and stations, train operation, working status of equipment, and energy consumption. Data center collects and processes these data and feeds results back to the command center for the command center to understand the operation of the whole line network, providing data support and services for train scheduling, equipment operation and maintenance, emergency handling, and other operational decisions. As shown in Fig. 5, the data acquisition layer, the data transmission layer, and the data application layer are included. The data acquisition layer is the data source of the information platform and the terminal layer of the whole system, including radio frequency identification (RFID) tag and reader, camera, and various wireless sensors, etc. Its function is to collect real-time data in rail transit. The attributes and features of these objects are converted into the form that can be recognized and processed by the computer system and then transmitted to the transmission layer. The data transmission layer is responsible for intelligent processing and network transmission of collected data. Network transmission layer mainly includes Wi-Fi, ZigBee, video cable, and other wired and wireless communication networks. How to use ZigBee technology to realize mass data transmission is one of the key points to realize the functions of rail transit system platform. The role of application layer lies in unified management for the approval of platform services, maintenance of normal operation of rail transit system platform, and completion of intelligent video analysis and other tasks.

Based on the interaction between the DTs-based city and the physical city, scientific decisions can be put forward through continuous learning, deduction, and prediction of possible bottlenecks in urban operation. Therefore, a smart city that is constantly self-renewing and evolving can be established. Alam and El Saddik proposed a reference model of DTs architecture for Cyber Twins Physical System (C2PS) based on Cloud Computing and analyzed the key characteristics of the C2PS, which helps to identify different levels of basic and hybrid computing interaction patterns. In addition, a C2PS intelligent interactive controller was designed using Bayesian belief network, so that the system can dynamically consider the current environment [11]. By applying DTs technology to smart city construction, Saddik found that DTs helps monitor and optimize the functions of all physical entities and provides continuous traffic information feedback to improve people's life quality [12].

In the interaction scenarios of various fields in the future smart city, different functional units and building units are assembled, so that people can diversify the use of micro-functions and building units. For example, intelligent sensing sidewalks, intelligent bicycle paths, and intelligent parks are common in cities, making the construction of interactive scenes more humanized. This is also an important way to promote the value increment of urban space by means of smart city. Based on the urban traffic DTs, the traffic flow data and real-time transmission data of road facilities can be analyzed and predicted, and the real-time optimization control of traffic lights can be carried out. Based on the historical data of urban traffic, it can predict the traffic situation of a certain area in the next 10 min to an hour and help the traffic management department to formulate emergency plans and implement traffic diversion measures before traffic congestion.

Relevant scholars use DTs technology to intelligently predict the entire life cycle of transportation infrastructure. Hou et al. proposed a link prediction method based on similarity to assess the life cycle of smart cities, which only relied on limited known data to estimate missing unit data. It was found that this calculation method could predict the life cycle of smart cities [13]. In view of the

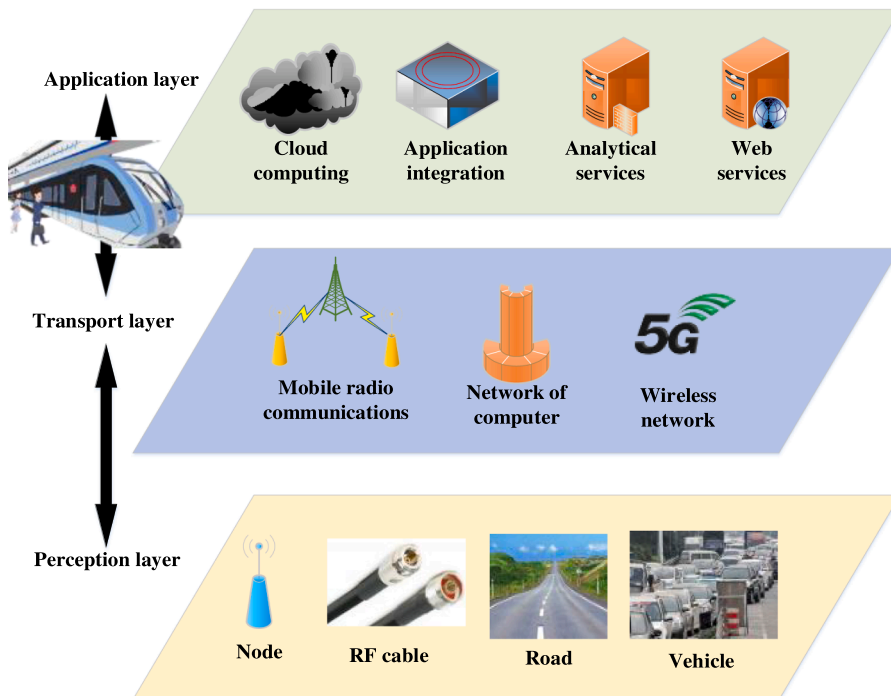


Fig. 5. Schematic diagram of rail transit real-time data processing framework based on Internet of Things and Big data analysis.

problems faced by traditional manufacturing industry, Li et al. proposed a DTs-driven sustainability evaluation information architecture oriented to the dynamic evolution of the whole life cycle in intelligent manufacturing and verified its effectiveness [14]. White et al. found that as the precise and powerful BIM in smart cities are more closely combined with the big data generated by Internet of Things sensors, and the massive data in smart cities become more accurate and transparent. The public and open DTs model makes the planning of smart city industrialization clearer [15].

Researchers in related fields have proved that using DTs technology can improve the accuracy of life cycle prediction. DTs technology is to make full use of physical model, sensor update, operation history, and other data, integrate multi-discipline, multi-physical quantity, multi-scale, multi-probability simulation process, and complete the mapping in virtual space, so that reflecting the full life cycle of the corresponding urban entity process. In the follow-up life cycle prediction of transportation infrastructure, deep learning technology can be further integrated to build a neural network-based life cycle prediction model based on DTs technology in transportation infrastructure, as shown in Fig. 6.

In the prediction of the life cycle of the traffic infrastructure, it can build "physical" and "virtual" data connection channels, the physical world, and the virtual world using DTs technology. In addition, it can create a digital virtual entity using the DTs technology with the help of historical data, real-time data, and algorithm model. The main characteristics of DTs technology include digitalization, interaction, foreknowledge, and foresight. The digitalization requires the representation of physical objects as digital models that computers and networks can recognize. Interaction is mainly real-time dynamic interaction between virtual objects and physical objects, and the Internet of Things is the core technology to realize it. One of the responsibilities of the virtual world is to predict and optimize physical world, so the instructions need to be transmitted to the physical world. The new state of the physical world needs to be transmitted in real time to the virtual world as new initial values and new boundary conditions for the digital world. Foreknowledge and foresight are predictions. The foreknowledge refers to the dynamic prediction of the physical world by using simulation technology, which requires digital objects not only to express the geometric shapes of the physical world, but also to incorporate physical laws and mechanisms into the digital model, which is a full-cycle and full-field dynamic simulation. The foresight is to use industrial big data and machine learning technology to foresee the future, which can surpass human deterministic knowledge of the physical world.

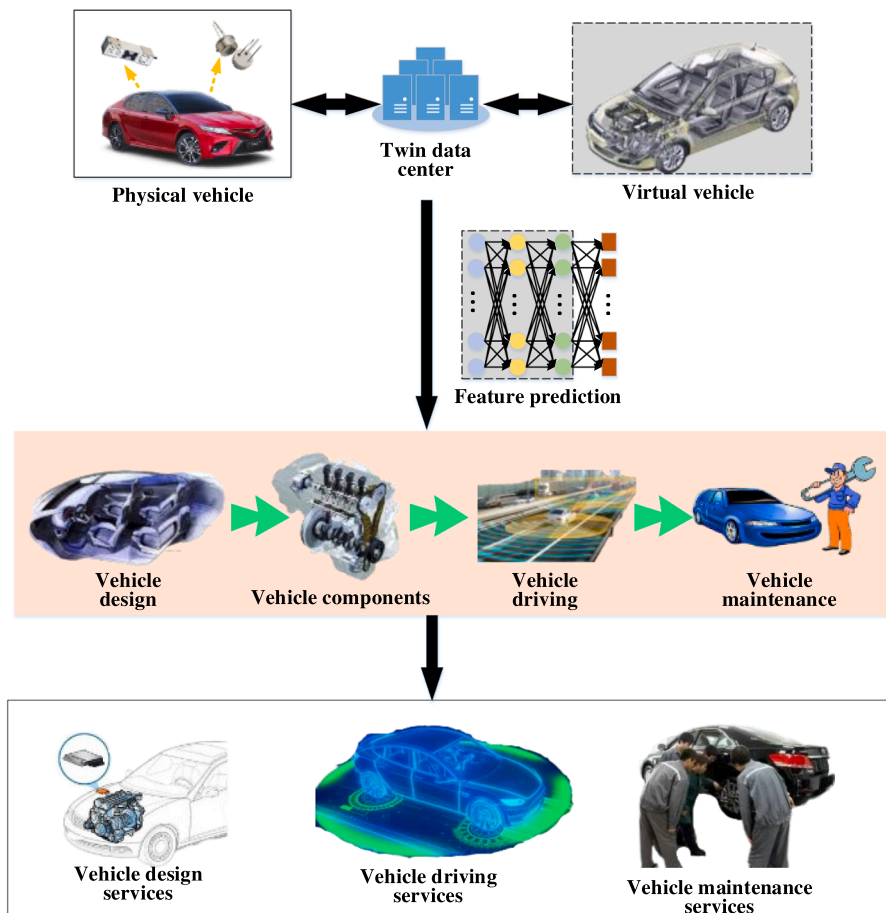


Fig. 6. Schematic diagram of the life cycle prediction model of the digital twin of the transportation infrastructure based on the neural network.

3.2.3. Prospective application of the combination of transportation infrastructure and New Media

Nowadays, the rapid development of digital media technology and the continuous updating of hardware such as smart phones and computers, mark the emergence of the New Media era with intelligent technology as the core. The combination of transportation infrastructure and New Media can solve the information island, release data value, optimize the allocation of transportation facilities resources, and promote the construction of smart cities.

In recent years, relevant scholars have conducted extensive research on the development of New Media. Gomez-barquero et al. used the 5G mobile broadband networks to transform different verticals including media broadcasting, and 5G systems could support and interoperate with non-3G Partnership Project (3GPP) access technologies, allowing multiple access technologies to be used simultaneously for one or more services, creating a new convergence scenario [16]. Marai et al. considered the current physical vulnerability of urban transport and used New Media to collect real-time data for assessment, which could help urban transport decision makers to analyze related potential risks for data-driven infrastructure management during extreme weather events [17].

Relevant scholars believe that the combination of transportation infrastructure and New Media mainly contains the following characteristics: interactivity, science and technology, and universality. Different from other arts, the most important feature of New Media art is its high participation and interaction. New Media art transmits its own creativity to participants through the media, and participants respond to the information received. Through the overall design and control of the space, carrier, and environment where the New Media art is located, viewers can participate in the works. With the rapid development of science and technology, the carrier of New Media art will be ever-changing, but they cannot leave the support of new science and technology. The New Media art also shows the possibility of more carrier changes, to make the imagination of designer become reality through the support of science and technology. Universality is mainly analyzed from the perspective of popularization of its application fields. The emergence of New Media art has broken traditional art forms, not only making more art works that are hard to contact enter the public's life in a more tangible form, but also making art creation itself more and more routine style and making the works more acceptable.

New Media is further combined with transportation infrastructure. In the construction of transportation infrastructure, the ITS is the key construction goal of developed countries. For example, they pay attention to timely collection, analysis, and feedback of information, and focuses on strengthening coordination and cooperation among various functional departments to achieve intelligent construction of traffic. Sodhro et al. proposed an adaptive green (i.e., energy-saving) algorithm based on 5G to optimize energy and reliability according to received signal strength indicator (RSSI) and packet loss ratio. All transmitted packets must be carried out under the constraint of minimum transmitted power with high reliability in a particular active time slot. Experimental results showed that this method significantly obtains 55% green and 41% reliable [18]. Arthurs et al. analyzed the relationship between cloud computing and intelligent transportation, argued that cloud computing near the network edge of vehicles and their sensors can provide

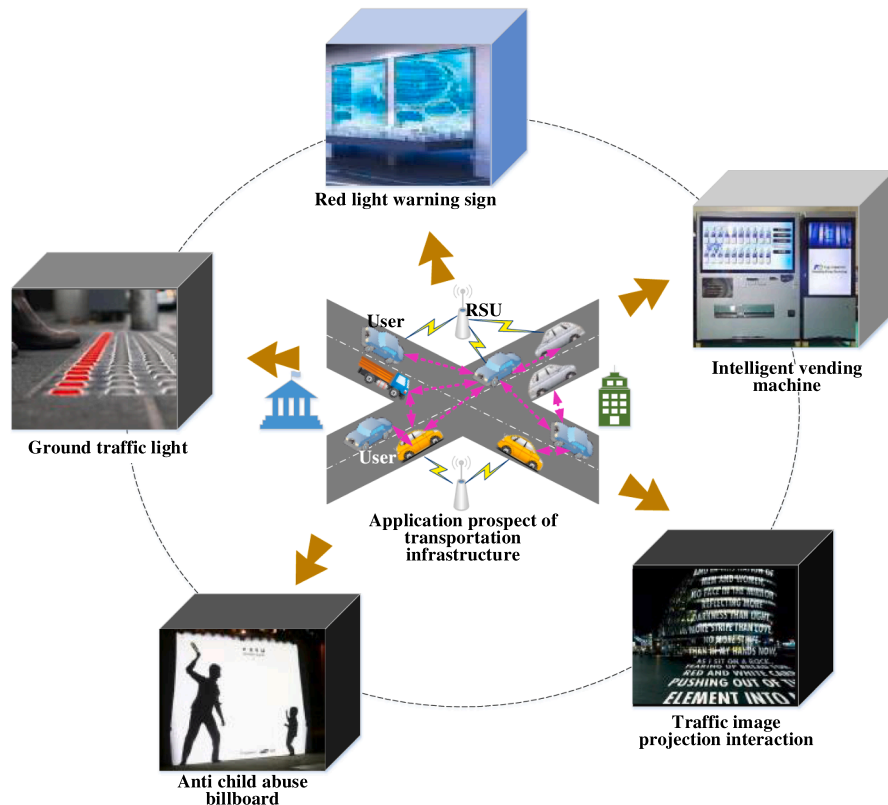


Fig. 7. A case application diagram of the combination of New Media and urban transportation infrastructure.

solutions for delay and bandwidth limitation, but the high mobility of vehicles and the heterogeneity of infrastructure still need to be solved [19].

Nowadays, the intelligent transportation has gradually matured. Cities with rapid economic development have built relatively complete intelligent transportation infrastructure systems, including comprehensive intelligent infrastructure such as road monitoring systems and traffic management systems, especially in the traffic information infrastructure, such as advertising light box, LED information touch screen. New Media has broken the traditional single mode of information transmission, and audiences can feedback information through the Internet and other ways, forming a two-way interactive environment and greatly improving the social participation of people. Relevant scholars have also studied the development of traffic intelligence. Pedersen et al. studied an autonomous ship system based on autonomous navigation system and DTs technology, which combined sensors and New Media technology to generate test scenarios and evaluate the safety, consistency, and efficiency of ship navigation [20]. The current technology already can support the development of New Media art, and infrastructure in the field of more and more design has been combined with New Media art. DTs combined with New Media technology would collect data in real time, synchronize traffic operation visualization, and provide experimental space for traffic model deduction to complete data-driven decisions. The most important aspect of transportation infrastructure is its interaction with urban residents, which is the most significant advantage of New Media art compared with traditional art. Under the trend of intelligent environment, the combination of transportation infrastructure construction and New Media art is a general direction of development in the future, whose specific case on is shown in Fig. 7.

The traffic light warning sign is an example of using digital and intelligent technologies to combine transportation infrastructure with New Media. When passers-by running red lights, light boxes will issue the voice of tires when the car braking and capture the people's frightened facial expressions then upload to the system through the network technology. Then, the light box is banned, and the red light will be immediately updated to the captured picture. This is a way to warn passers-by against running red lights [21]. In addition, the scrolling mapping of fonts on the road can break the traditional art form and bring great visual impact to viewers. Through the "dematerialized" expression in New Media art design, personal thoughts can be expressed in the form of images, sounds, and words. In this way, the output carrier such as ordinary billboard and LED screen is abandoned, and it is displayed in a direct and open form. Another example is vending machines in road network areas. The difference is that it does not need to drop a coin, but just stand in front of the machine with a big smile, and the machine will deliver a 550 mL bottle of mineral water for free. This seemingly ordinary transportation infrastructure brings a little surprise into the user's life by adding Human-Computer Interaction and New Media [22]. With the various New Media technologies, traffic facilities can provide residents with data and information they need at any time. In the design of transportation facilities system, New Media technology is combined to enhance public participation and social cohesion. The main goal is to build "Big System, Big Platform, and Big Data", promote the interconnection of traffic control system and data exchange platform, combine social media data, traffic equipment perception data, and system data, and create a double closed-loop traffic control system of social system and physical information system.

In the past, traffic control would build the data interaction architecture of foreground and background. The foreground collects data of users, and the background is responsible for data management. When the scale of data gradually expands, the contradiction between the foreground and background will become complicated. Different foregrounds will correspond to different types of data, and the background will be more difficult to handle in terms of data support and control. New Media is an emerging field, the key lies in its strong participation and interaction, but there is no clear and standardized definition of it until now. Urban traffic facilities are essential for an urban space, so the information conveyed by these facilities must be easily received by the public. Meanwhile, most of the traffic facilities should have design requirements such as scale, color, and font. Therefore, the combination of the two should consider the standards and requirements of the transportation facilities themselves, but the processing of artistic elements will be greatly limited on this basis. The projection technology, holographic visual effect, and other technologies have high requirements for the site selection of works, such as that the weather and lighting of the surrounding environment will have an impact on the final effect. Therefore, with the support of New Media, there are still many difficulties in the combination of urban transportation facilities and New Media, which is also the direction for scholars in related fields to explore urban development in the future.

4. Conclusion

Based on smart cities, it classifies and predicts the status of intelligent development of transportation infrastructure. The applications of transportation infrastructure based on AI technology and DTs technology are analyzed to predict the intelligent development of transportation infrastructure and provide new development opportunities for the construction of smart city transportation system. In addition, it analyzes and summarizes the research in this field, and helps the operation and management department of intelligent transportation infrastructure to monitor the operation status of transportation infrastructure in real time, find problems in time, and maintain them, reducing inefficiency and losses.

In the following research, it will further explore the popularization and promotion of the smart transportation DTs system, so that it can be truly implemented and provide services for the construction of the final smart city. At present, the biggest challenge of AI and DTs is the application landing and how to make more industries and scenarios really use it. AI and DTs should be combined with the professional knowledge involved in the application scenario to maximize the value of data.

CRedit authorship contribution statement

Jingyi Wu: Investigation, Data curation, Visualization, Writing – original draft. **Xiao Wang:** Supervision, Project administration. **Yukun Dang:** Methodology, Supervision, Validation. **Zhihan Lv:** Methodology, Supervision, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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